

Rethinking case attraction on Ancient Greek infinitive clauses

Caio B. A. Geraldес*

<caio.geraldes@usp.br>

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1 Introduction

Nominal predicates and adjuncts modifying subjects of complement infinitive clauses in Ancient Greek (AG) usually agree in case with their domain local subject, which has morphological accusative case assigned in the *Accusativus cum Infinitivo* (Acl) construction, e.g. (1-a). The same occurs when the subject is not overtly expressed, e.g. (1-b).

- (1) a. ἐγὼ οὖν, ὦ Πρωταγόρα, εἰς ταῦτα ἀποβλέπων οὐχ ἡγοῦμαι **διδασκτὸν** εἶναι **ἀρετῆν**
I therefore, Protagoras, in view of such, do not believe virtue to be teachable. (Plat. *Prot.* 320b)
- b. ἀλλ' ἔμοιγε, ἔφη, ὦ Σώκρατες, **διδασκτὸν** εἶναι δοκεῖ.
For my part, Socrates, he said, I think [wisdom] it is teachable. (Plat. *Euthyd.* 282c)

This type of agreement is the default in AG infinitive clauses and seemingly applies all the general rules of predicate and adjunct concord in local domains in AG.

However, when the subject of complement the infinitive is empty and refers to the same discourse entity expressed by a constituent of the matrix clause, other agreement resolutions occur. When the subject of the infinitive is the same as that of the matrix clause, the predicate or adjunct displays the nominative case agreeing in case with the constituent of the non-local domain, e.g. (2). This

*Fundação de Amparo à Pesquisa do Estado de São Paulo.

agreement does not seem to be optional, as there is little to no evidence of sentences with the modifier in any case other than nominative.

- (2) ... ὥς ἐγὼ μὲν οὐχ ὁμολογήσω ἄκλητος ἦκειν, ἀλλ' ὑπὸ σοῦ κεκλημένος.
... for I shall not admit that I came unasked, but only on your invitation.
(Plat. *Symp.* 174c)

When other constituents besides the matrix' subject "fight" for the control of the modifier, so to speak, a couple of exceptions surface. If there is an emphatic pronoun in the infinitive clause, it takes accusative and it's modifiers agree with it, (3).

- (3) ταῦτ' οὖν πάνυ μὲν οὖν οὐκ οἶομαι οὕτως ἔχειν, ἀλλ' ἐμὲ φαῦλον εἶναι
ζητητήν
Now I really do not think that this can be the case, but rather that I am a poor hand at inquiring. (Plat. *Charm.* 175e)

The other exception is restricted to an idiosyncratic personal construction of the verb δοκέω, in which the subject and indirect object both refer to the same entity, on which both the subject's nominative and object's dative are available for agreement, (4). The accusative case is never attested — but bear in mind that this construction is rarely attested overall, so that this lack of attestation may be accidental.

- (4) a. δοκῶ γάρ μοι ἀδύνατος εἶναι
For I believe myself to be incapable. (Plat. *Resp.* 368b)
b. ὅμοιος ἄν μοι δοκῶ εἶναι τῷ περιόντι ἱατρῷ καὶ ἐπισκοποῦντι
τοὺς κάμνοντας, εἰδότι δὲ οὐδὲν ὅ τι συμφέρει τοῖς κάμνουσιν.
I think I should be like that doctor who goes round visiting the sick,
but has no knowledge of the right way to treat them. (Xen. *Econ.* 15.9)

The examples (2) and (4), particularly (4-b), may be called *case attraction*, but the term is usually reserved to the next type of case concord pattern. When the subject of the complement infinitive refers to the same entity as an oblique argument of the matrix clause, its predicates or adjuncts may either bear accusative, (5-a), or agree in case with their co-referential controller, (5-b), the latter being what is generally called *case attraction*.

- (5) a. συμβουλεύει τῷ Ξενοφῶντι ἐλθόντα εἰς Δελφοὺς ἀνακοινῶσαι τῷ
θεῷ περὶ τῆς πορείας.
He advises Xenophon to go to Delphi and ask the god about the travel. (Xen. *Anab.* 3.1.5)

- b. ἀφῆκε **μοι ἐλθόντι** πρὸς ὑμᾶς λέγειν τᾰληθῆ.
He allowed me to go and speak the truth in front of you. (Xen. *Hell.* 6.1.13)

By naming it as *case attraction*, grammarians and philologists have assumed, since Lancelot (1658, 1709) extended the term used by Sánchez de las Brozas (1587) for case attraction of relative pronouns, that (5-b) is some type of deviation from the general rule of agreement/concord in AG (and Classical Latin, for that matter). Since then, however, the cause for the variation of constructions with (5-b) and without (5-a) case attraction remained largely unexplained. Authors have only tentatively drawn contexts in which a certain option is more likely to occur, a handful of times assuming that such correlation between context and resolution was causal, often due to *emphasis* and *focus*, without any empirical evidence for the relation between *emphasis* and attraction.¹ Other works have simply asserted its existence.²

When it came the time for general linguists to analyse AG case agreement patterns shown in the previous examples, their approach was rather different, as they bundled together what a philologist would call “normal” case agreement and case attraction. Since Andrews (1971), Lakoff (1970), and Quicoli (1982), the works dealing with AG case agreement patterns have been trying to provide a structural explanation for *how* non-local case concord is configured in the syntax. Most recently, Sevdali (2013a,b) (drawing part of their analysis from the works of Tantalou 2003 and Spyropoulos 2005) argued that AG infinitive clauses may evaluate their Person head locally or externally. For them to be locally evaluated, a “discourse prominent” element, not present in the infinitive clause, must be supplied in the form of an optional accusative *pro*, which would be the case of (5-a) sentences. Otherwise, the caseless PRO would probe for case until it got it from the matrix constituent which controls it, thus the case agreement in (5-b) would surface. The cause for case attraction, under this model, would be the lack of a “discourse prominent” element available to be represented by an accusative *pro* in the infinitive clause. Similarly to the philologists’ “emphasis” explanation, however, the “discourse prominence” does not come with empirical evidence or clear definition.

The issue of “emphasis”, “focus” and “discourse prominence” is not just a shortcoming or unsatisfactory limit of scope in the works on case attraction. It is also a measuring issue: there are no metrics for *emphaticness* or *prominence* and the approximating techniques available for studies on contemporary languages,

¹Buttmann (1826, p. 462), Hocheder (1833), Kühner and Gerth (1898, II pp. 24ff.), Cooper and Krüger (1997, p. 774), and Cooper and Krüger (2002, pp. 2500–1).

²Goodwin (1900, p. 203), Stahl (1907, p. 643), Schwyzler (1950, II p. 368), Smyth (1956, p. 278, p. 440), Rijksbaron (2002, p. 105), Napoli (2014), Boas et al. (2019, p. 587).

namely the experimental techniques ranging from interviews and questionnaires all the way to eye-tracking and reaction timing, are not feasible when it comes to *corpus* languages such as AG.

2 A stochastic approach

Using data from literary sources of Classical Greek, including oratory speeches, drama, historiography and philosophical dialogues from Attic and Ionic sources, I provide a data driven quantitative analysis of the contexts in which case attraction is a possible agreement / concord resolution. The addition of Ionic sources is due to the fact that it has been assumed that case attraction is more common if not the rule in the Attic dialect (e.g. Buttman 1826, *passim*. and Cooper and Krüger 1997, *ad loc.*). The data has been collected and annotated in a combination of manual and computational methods using the Diorisis Ancient Greek Corpus (Vatri and McGillivray 2018).

A quantitative analysis of case attraction requires caution in the methodological approach, as semantic and pragmatic features are often latent, i.e. not explicitly present in the morpho-phonological level, and the use of proxies such as word classes, constituent distance and word order may hinder the quality of the results and the causal inference built upon them. As such, our analysis will rely on the causality analysis (e.g. Pearl 2009) and Bayesian modeling (e.g. McElreath 2020), which I argue could enhance the results of data driven linguistic research by including at the quantitative analysis the qualitative knowledge built on the Ancient Greek and general linguistics. The analysis must be thus twofold. Firstly, it assess how the linguistic and extralinguistic factors could reasonably be causally linked with case attraction, so as to inform the statistical modeling and tell what effects are possible to estimate from the data. Later, a general linear model is build as to adequately estimate the direct effects of semantic and pragmatic factors on case attraction and the interactions between linguistic and extralinguistic variables.

3 A causal model for case attraction

Case attraction consists in a covariance of a specific morphological feature (case) between a controller (the oblique argument of a certain matrix verb) and a target (the predicative or attributive expression modifying the logical subject of the infinitive verb). Thus, it falls into the general concept of *agreement*, in Corbett's framework (2006, pp. 4–7).

By *agreement*, we mean the general class of phenomena by which morphological features covary across a pair of tokens with a particular direction, i.e.: in which there are a controller and a target. Even there are indeed, as Norris (2014, 2017) and Polinsky (2016) argue, different mechanisms behind subject-verb agreement, DP-internal concord, and predicate case concord — being case attraction an instance of the last one —, our use of Corbett’s framework will remain unaltered, as the differences observed between DP-internal concord and subject-verb agreement in the authors defending the contrasts are irrelevant for our analysis.

By assuming that case attraction (*A*) is a sort of *non-canonical agreement/concord*, it must be expected that factors known cross-linguistically to trigger these *phenomena* also affect *A*. Drawing a list of conditioning factors for *non-canonical agreement* from Corbett (2006, pp. 176–205), we should thus expect that the following factors would change the likelihood of *A*. 1. thematic role (Θ) and animacy (*An*); 2. grammatical relation (*GR*) and case (*K*); and 3. communicative function (*CF*), and precedence (*Prec*).

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A Criteria for Canonical Agreement

The following list summarizes the criteria for canonical agreement as discussed in section 1.4 of Corbett (2006, pp. 8ff.). The underlines indicate what position in the hierarchy case attraction arguably occupies.

- Controllers:
 - C-1: controller present > controller absent
 - C-2: controller has overt expression of agreement features > controller has covert expression of agreement features
 - * “A canonical controller marks at least as many distinctions as the target”
 - C-3: consistent controller > hybrid controller
 - C-4: controller’s part of speech is irrelevant > is relevant (given the domain)
- Targets:
 - C-5: inflectional marking (affix) > clitic > free word
 - C-6: obligatory > optional
 - C-7: regular > suppletive
 - C-8: alliterative > opaque
 - C-9: productive marking of agreement > sporadic marking
 - C-10: target always agree > target agrees only when controller is absent
 - C-11: target agrees with a single controller > agrees with more than one controller
 - C-12: target has no choice of controller > target has choice of controller (is ‘trigger-happy’)
 - C-13: target’s part of speech is irrelevant > is relevant (given the domain)
- Domains:
 - C-14: asymmetric > symmetric
 - C-15: local domain > non-local domain
 - C-16: domain is one of a set > single domain
- Features:
 - C-17: feature is lexical > non-lexical
 - C-18: features have matching values > non-matching values
 - C-19: no choice of feature value > choice of value
- Conditions:
 - C-20: no conditions > conditions